

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 24)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Table 1 shows climate data for two sites where a farmer is considering growing maize, a frost-sensitive crop that needs a thermal growing season of at least 120 days.

Table 1

Feature	Site A: valley floor	Site B: south-facing slope
Frost-free period / days	110	150
Days with mean temperature above 5 °C	160	175
Mean summer rainfall / mm	180	150
Gradient / %	1	8

0 1

. 1

Use Table 1 to explain which site is more suitable for maize.

[3 marks]

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0 1

. 2

Describe **three** ways in which temperature can be controlled to increase crop productivity.

[3 marks]

1

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2

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3

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0 1

. 3

Explain why keeping livestock warm increases the amount of food converted to growth.

[2 marks]

.....

0 1 . 4

Explain how day length affects livestock production and how it can be manipulated.

[2 marks]

10

0 2 . 1

Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

- A Its mass
- B Its magnetic field
- C Its distance from the Sun
- D Its speed of rotation

☐

☐

☐

☐

0 2 . 2

Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

0 2 . 3

Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

5

03 . 1

Explain why an electronic meter used to measure dissolved oxygen must be calibrated, and what should be done if the reading cannot be adjusted.

[2 marks]

03 . 2

Give **three** types of indirect evidence of the presence of an animal and the information each can provide.

[3 marks]

1

2

3

03 . 3

Explain how auditory monitoring can be used to study bats.

[2 marks]

03 . 4

Explain **three** assumptions of the mark-release-recapture method that may not be met.

[3 marks]

10

0 4 . 1

Compare selective breeding, asexual reproduction and cross-breeding as methods of improving livestock or crops.

[6 marks]

0 4 . 2

Explain how artificial insemination and embryo transfer increase the rate at which desirable characteristics spread through a livestock population.

[4 marks]

10

05 . 1

Describe how each of the following abiotic factors can be managed to increase biodiversity in a nature reserve.

[6 marks]

Light

Dissolved oxygen

Nutrients

05 . 2

Explain how biotic factors can be managed to increase the survival of a species. Give **four** examples.

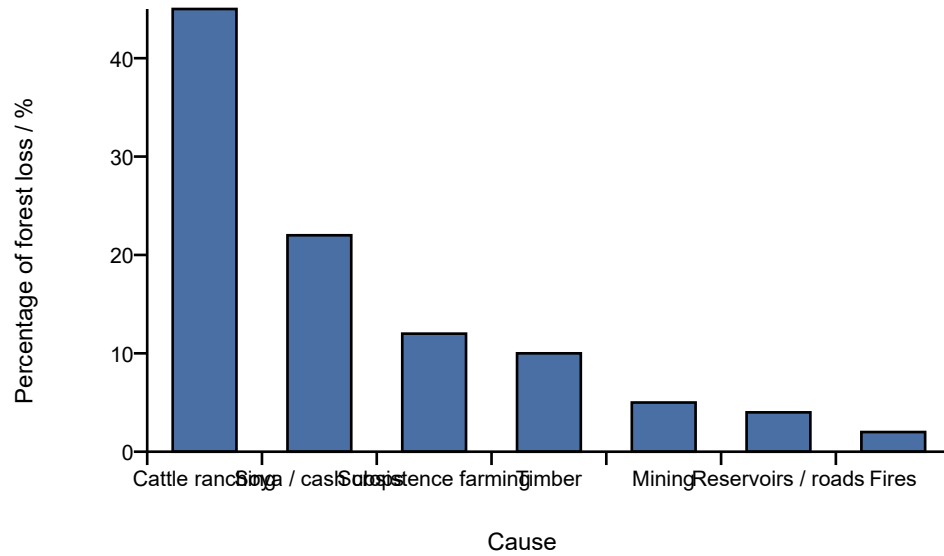
[4 marks]

10

06

Figure 1 shows the main causes of tropical rainforest loss in a South American country between 2000 and 2020.

Figure 1



06

1

Explain why traditional 'slash and burn' subsistence agriculture was sustainable but has become a threat.

[3 marks]

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06

2

Explain why clearance for cattle ranching or cash crops is usually unsustainable.

[2 marks]

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.....

.....

06 . 3

Describe how each of the following threatens rainforest.

[3 marks]

Mineral extraction

HEP reservoirs

Climate change

06 . 4

Describe how a debt-for-nature swap works.

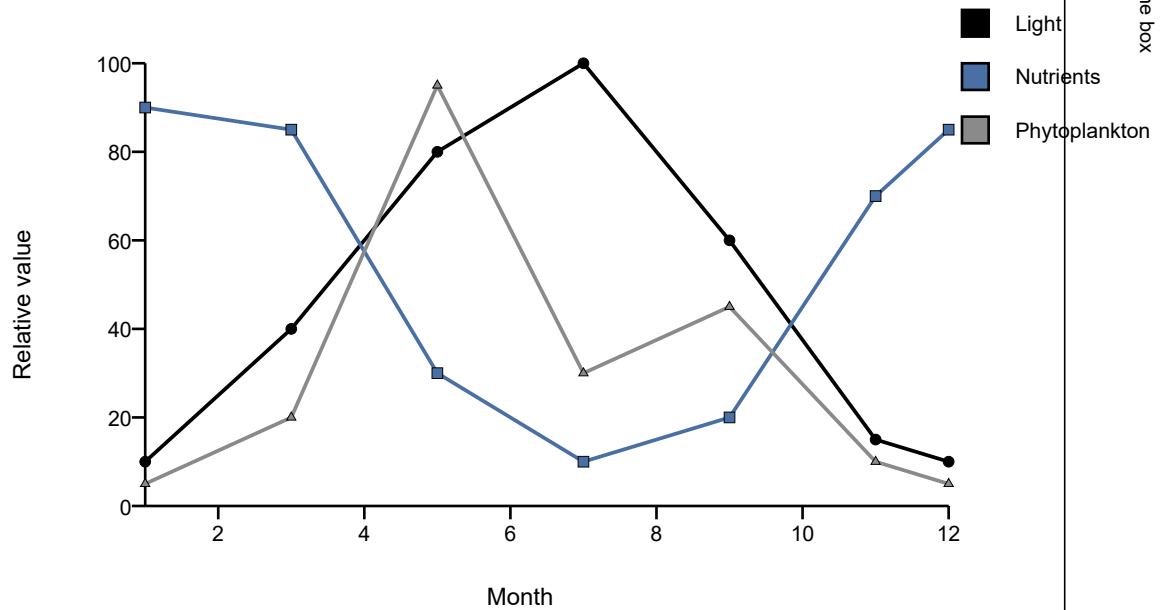
[2 marks]

10

07

Figure 1 shows the seasonal changes in light intensity, nutrient concentration and phytoplankton biomass in the surface water of a temperate ocean in the northern hemisphere.

Figure 1



07

1 Use Figure 1 to explain the spring peak and summer decline in phytoplankton biomass.

[4 marks]

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07

2 Explain why tropical open oceans have low productivity all year despite high light and temperature.

[2 marks]

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07 . 3

Explain why areas of upwelling and coastal waters are the most productive parts of the sea.

[3 marks]

07 . 4

Explain why phosphate is often the limiting nutrient in the sea.

[1 mark]

10

08 . 1

Describe how agriculture alters the hydrological cycle.

[3 marks]

08 . 2

Explain how each of the following consumer choices can reduce the environmental impact of food production.

[4 marks]

Seasonal / local food

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Organic food

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08 . 3

Explain why post-war European governments introduced grants and guaranteed prices, and describe **one** environmental consequence.

[3 marks]

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10

09

Table 1 lists some techniques used in ecological monitoring.

09 . 1

Complete Table 1 by giving one application and one limitation of each technique.

[6 marks]

Table 1

Technique	Application	Limitation	
Image databases of unique markings			
eDNA sampling			
Geolocator tags			

.....

.....

09 . 2

Explain how satellite radar (microwave) sensors are useful for monitoring oil pollution and sea conditions.

[2 marks]

09 . 3

Give **two** types of indirect evidence of an animal's presence and the information each can give.

[2 marks]

1

2

10

Discuss the reasons why the conservation of biodiversity is important for human society.

ESP/7447/2/SET24

Extra space

10 . 2 State what is meant by **biodiversity**.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

[illegible]

END OF QUESTIONS